RHCSA CHALLENGE 2

SA –2

Submitted By

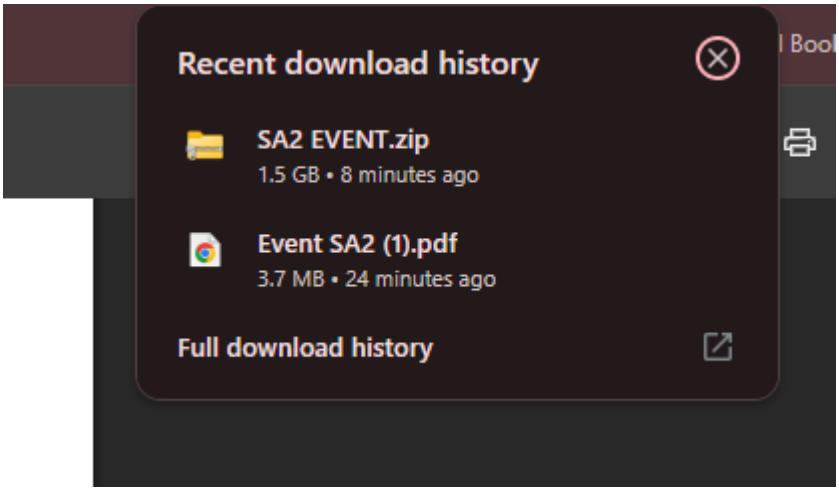
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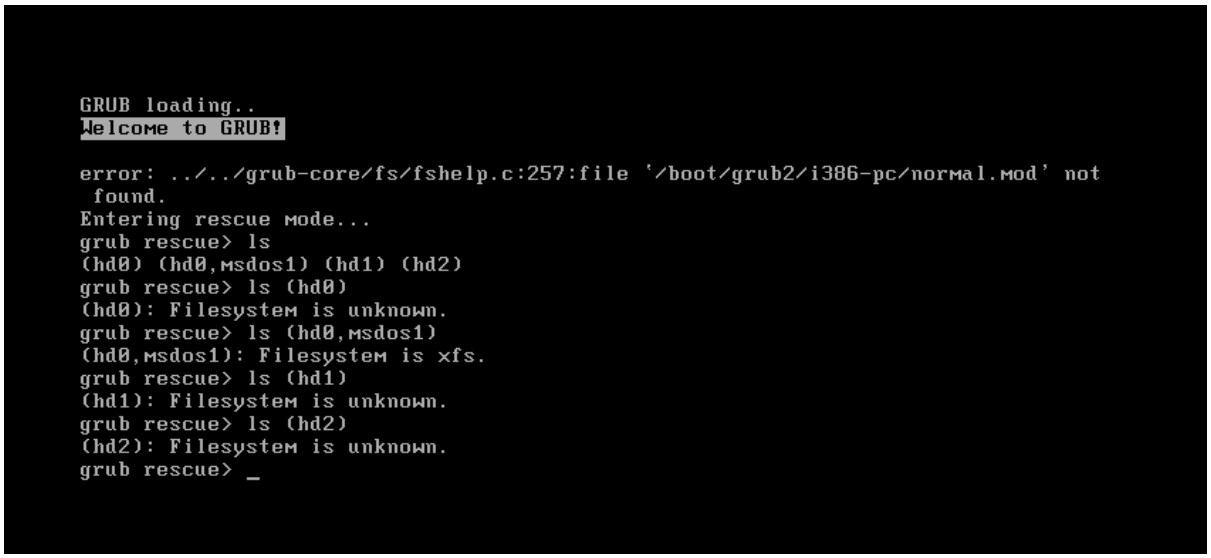
STEP 1: Download and Start the Operating System.

TASK 1

- Dowloaded the OS from Google Drive



- Setting up the os
- When the computer started, GRUB showed an error saying that /boot/grub2/i386-pc/normal.mod was not found and entered rescue mode. This meant GRUB itself was starting, but it could not load its normal module.



- From grub rescue, I checked the available disks and partitions. The important partition was (hd0,msdos1).

```
grub rescue> ls hd2/

error: ../../grub-core/fs/fshelp.c:281:invalid file name 'hd2/'.
grub rescue> ls (hd2)/
error: ../../grub-core/kern/fs.c:121:unknown filesystem.
grub rescue> ls (hd0,msdos1)/
./ ../ dev/ proc/ run/ sys/ tmp/ etc/ root/ var/ usr/ bin sbin lib lib64 afs/ bo
ot/ home/ media/ mnt/ opt/ srv/
grub rescue>
```

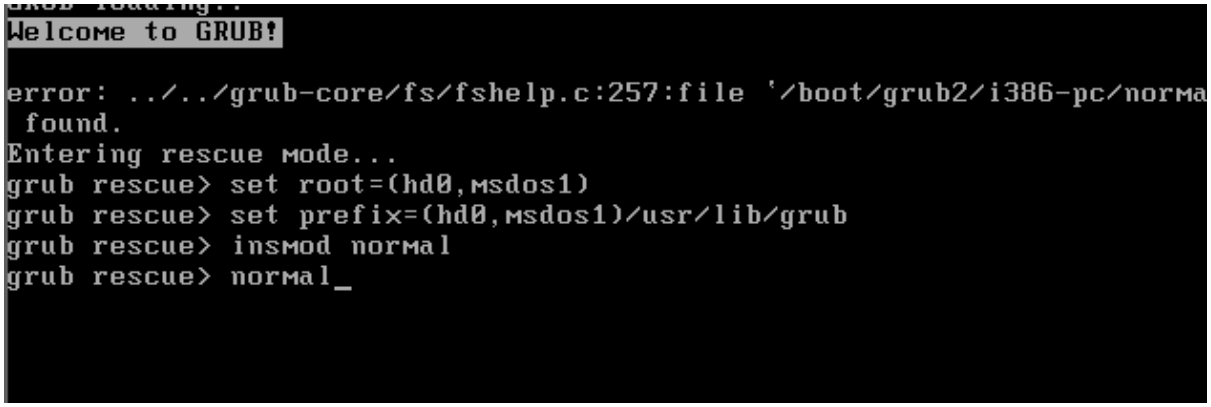
- Finding the kernel and GRUB files.

```
grub rescue> ls (hd0)/
error: ../../grub-core/kern/fs.c:121:unknown filesystem.
grub rescue> ls /
./ ../ dev/ proc/ run/ sys/ tmp/ etc/ root/ var/ usr/ bin sbin lib lib64 afs/ bo
ot/ home/ media/ mnt/ opt/ srv/
grub rescue> ls /boot/
./ ../ efi/ loader/ vmlinuz-5.14.0-611.5.1.el9_7.x86_64 System.map-5.14.0-611.5.
1.el9_7.x86_64 config-5.14.0-611.5.1.el9_7.x86_64 .vmlinuz-5.14.0-611.5.1.el9_7.
x86_64.hmac symvers-5.14.0-611.5.1.el9_7.x86_64.gz initramfs-5.14.0-611.5.1.el9_
7.x86_64.img vmlinuz-0-rescue-22611a6ea130461e97f2aa80d137ec20 .vmlinuz-0-rescue
-22611a6ea130461e97f2aa80d137ec20.hmac initramfs-0-rescue-22611a6ea130461e97f2aa
80d137ec20.img initramfs-5.14.0-611.5.1.el9_7.x86_64kdump.img
grub rescue>
```

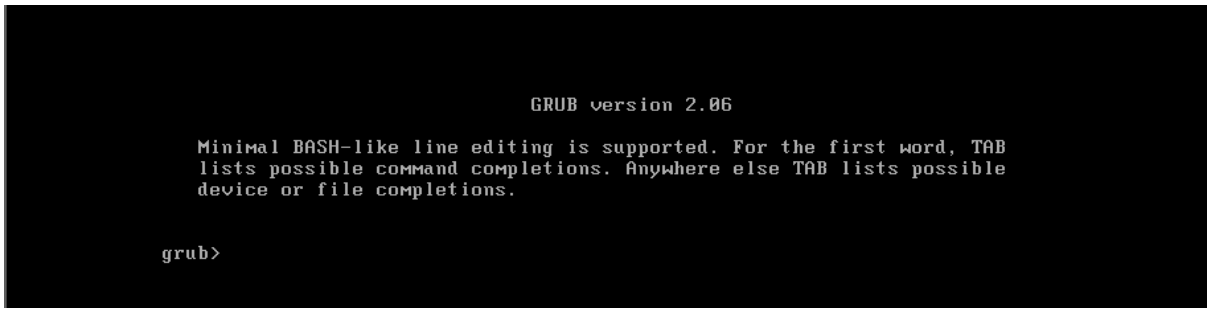
- Finding normal.mod under /usr/lib/grub/i386-pc

```
macbless.mod macho.mod mda_text.mod mdraid09.mod mdraid09_be.mod mdraid1x.mod me
mdisk.mod memrw.mod minicmd.mod minix.mod minix2.mod minix2_be.mod minix3.mod mi
nix3_be.mod minix_be.mod mmap.mod moddep.lst modinfo.sh morse.mod mpi.mod msdosp
art.mod mul_test.mod multiboot.mod multiboot2.mod nativedisk.mod net.mod newc.mo
d nilfs2.mod normal.mod ntfs.mod ntfscomp.mod ntldr.mod odc.mod offsetio.mod ohc
i.mod part_acorn.mod part_amiga.mod part_apple.mod part_bsd.mod part_dfly.mod pa
rt_dvh.mod part_gpt.mod part_msdos.mod part_plan.mod part_sun.mod part_sunpc.mod
 partmap.lst parttool.lst parttool.mod password.mod password_pbkdf2.mod pata.mod
 pbkdf2.mod pbkdf2_test.mod pci.mod pcidump.mod pgp.mod pkcs1_v15.mod plainmount
.mod plan9.mod play.mod png.mod priority_queue.mod probe.mod procfs.mod progress
.mod pxe.mod pxeboot.img pxechain.mod raid5rec.mod raid6rec.mod random.mod rdmsr
.mod read.mod reboot.mod regexp.mod reiserfs.mod relocator.mod romfs.mod scsi.mo
d search.mod search_fs_file.mod search_fs_uuid.mod search_label.mod sendkey.mod
serial.mod setjmp.mod setjmp_test.mod setpci.mod sfs.mod shift_test.mod signatur
e_test.mod sleep.mod sleep_test.mod smbios.mod spkmodem.mod squash4.mod strtoull
_test.mod syslinuxcfg.mod tar.mod terminal.lst terminal.mod terminfo.mod test.mo
d test_asn1.mod test_blockarg.mod testload.mod testspeed.mod tftp.mod tga.mod ti
me.mod tr.mod trig.mod true.mod truecrypt.mod udf.mod ufs1.mod ufs1_be.mod ufs2.
mod uhci.mod usb.mod usb_keyboard.mod usbms.mod usbserial_common.mod usbserial_f
tdi.mod usbserial_pl2303.mod usbserial_usbdebug.mod usbtest.mod vbe.mod version.
mod vga.mod vga_text.mod video.lst video.mod video_bochs.mod video_cirrus.mod vi
deo_colors.mod video_fb.mod videoinfo.mod videotest.mod videotest_checksum.mod w
rmsr.mod xfs.mod xnu.mod xnu_uuid.mod xnu_uuid_test.mod xzio.mod zfs.mod zfscryp
t.mod zfsinfo.mod zstd.mod
grub rescue> _
```

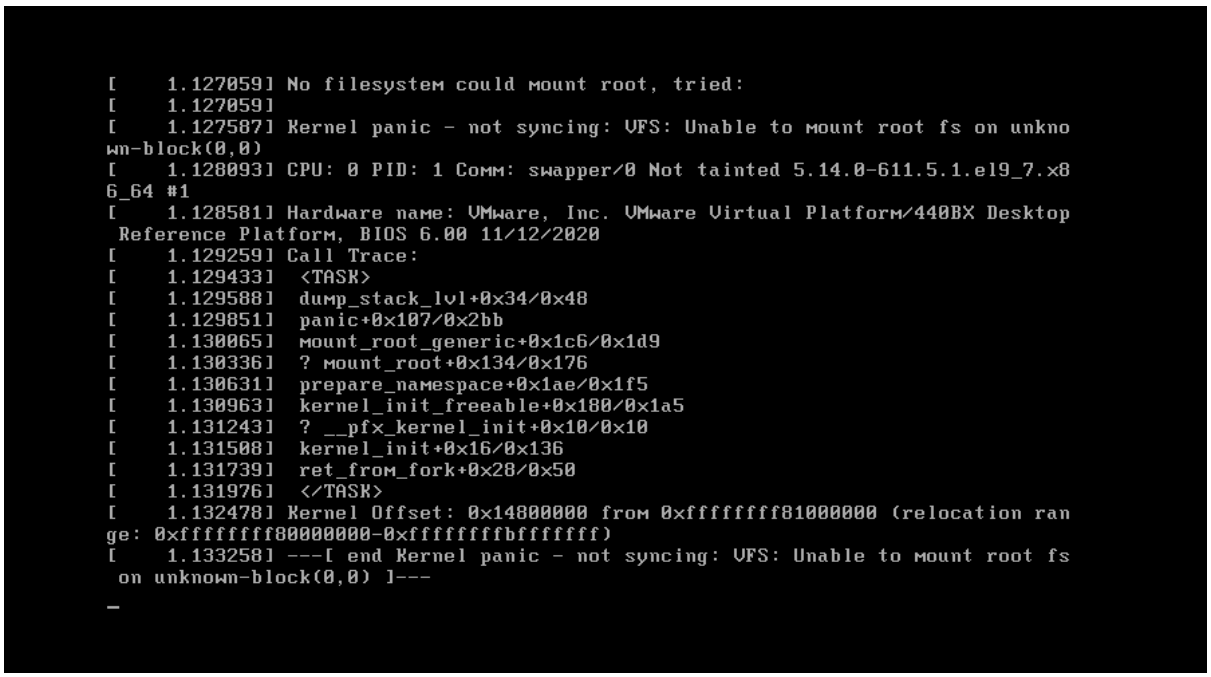
- Setting the GRUB prefix.



- Reaching the full GRUB command line



- Kernel panic from the wrong root device



- Reading /etc/fstab and finding the root UUID

```
(hd0) (hd0,msdos1) (hd1) (hd2)
grub> ls (hd0,msdos1)
        Partition hd0,msdos1: Filesystem type xfs, UUID
d1aa5bd2-a705-4dcf-b83a-307ce9263a6e - Partition start at 1024KiB - Total size
10484736KiB
grub> ls (hd0,msdos1)/etc/fstab
fstab
grub> cat (hd0,msdos1)/etc/fstab

#
# /etc/fstab
# Created by anaconda on Sat Jan  3 14:20:00 2026
#
# Accessible filesystems, by reference, are maintained under '/dev/disk/'.
# See man pages fstab(5), findfs(8), mount(8) and/or blkid(8) for more info.
#
# After editing this file, run 'systemctl daemon-reload' to update systemd
# units generated from this file.
#
UUID=d1aa5bd2-a705-4dcf-b83a-307ce9263a6e /                       xfs     defau
lts        0 0

grub> linux /boot/vmlinuz-5.14.0-611.5.1.el9_7.x86_64 root=UUID=d1aa5bd2-a705-4d

cf-b83a-307ce9263a6e ro
```

- Booting with the correct XFS root UUID.

```
not found.
grub>
grub> linux /boot/vmlinuz-5.14.0-611.5.1.el9_7.x86_64 root=UUID=d1aa5bd2-a705-4
grub> 3a-307ce9263a6e ro
grub>
grub> initrd /boot/initramfs-5.14.0-611.5.1.el9_7.x86_64.img
grub> boot_
```

- Astra Linux 9.7 boots successfully.

```
Astra login:
Astra login: root
Password:
Last login: Sun Aug 23 11:35:37 on tty1
[root@Astra ~]#
[root@Astra ~]#
[root@Astra ~]#
```

- Permanently Fixing GRUB.

```
[root@Astra ~]#
[root@Astra ~]# grub2-install /dev/sda
Installing for i386-pc platform.
Installation finished. No error reported.
[root@Astra ~]# grub2-mkconfig -o /boot/grub2/grub.cfg
Generating grub configuration file ...
Adding boot menu entry for UEFI Firmware Settings ...
done
[root@Astra ~]# ls -l /boot/grub2
total 36
drwxr-xr-x. 2 root root    25 Oct 11 05:08 fonts
-rw-------. 1 root root  6840 Oct 11 05:10 grub.cfg
-rw-r--r--. 1 root root  1024 Oct 11 05:08 grubenv
drwxr-xr-x. 2 root root 16384 Oct 11 05:09 i386-pc
drwxr-xr-x. 2 root root  4096 Oct 11 05:09 locale
[root@Astra ~]#
```

STEP 2: SYSTEM SECURITY, WEB SERVER & PERFORMANCE CONFIGURATION

Task 2.1 -Configure SSH Security

```
# This sshd was compiled with PATH=/usr/local/bin:/usr/bin:/usr/local/sb

# The strategy used for options in the default sshd_config shipped with
# OpenSSH is to specify options with their default value where
# possible, but leave them commented.  Uncommented options override the
# default value.

# To modify the system-wide sshd configuration, create a  *.conf  file u
#  /etc/ssh/sshd_config.d/  which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf

# If you want to change the port on a SELinux system, you have to tell
# SELinux about this change.
# semanage port -a -t ssh_port_t -p tcp #PORTNUMBER
#
#Port 2222
#AddressFamily any
#ListenAddress 0.0.0.0
#ListenAddress ::

#HostKey /etc/ssh/ssh_host_rsa_key
#HostKey /etc/ssh/ssh_host_ecdsa_key
#HostKey /etc/ssh/ssh_host_ed25519_key
```

- To Ensure the new port Works. We need to inform SELinux security polices that tcp 2222 port is an authorized port for ssh.

```
[root@Astra ~]# semanage port -a -t ssh_port_t -p tcp 2222
Port tcp/2222 already defined, modifying instead
[ 6272.569374] SELinux:  Converting 410 SID table entries...
[ 6272.588763] SELinux:  policy capability network_peer_controls=1
[ 6272.588929] SELinux:  policy capability open_perms=1
[ 6272.589036] SELinux:  policy capability extended_socket_class=1
[ 6272.589135] SELinux:  policy capability always_check_network=0
[ 6272.589231] SELinux:  policy capability cgroup_seclabel=1
[ 6272.589325] SELinux:  policy capability nnp_nosuid_transition=1
[ 6272.589478] SELinux:  policy capability genfs_seclabel_symlinks=1
```

- verifying the ssh port

```
[root@Astra ~]#
[root@Astra ~]# semanage port -l | grep ssh
ssh_port_t                     tcp      2222, 69, 22
[root@Astra ~]#
```

- To Check whether the SELinux is enabled or in enforce mode, type getenforce in the terminal.
- Newly Changed port (2222) should be updated to firewall as well Because By default firewall will only recognize the ssh rule as port 22, it should be permanent so that the configuration remains persistent after reboot.

```
[root@Astra ~]# sshd -t
[root@Astra ~]# firewall-cmd --permanent --add-port=2222/tcp
success
[root@Astra ~]# firewall-cmd --list-all | grep ssh
  services: cockpit dhcpv6-client http mountd nfs rpc-bind ssh
[root@Astra ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client http mountd nfs rpc-bind ssh
  ports:
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@Astra ~]# firewall-cmd --reload
success
[root@Astra ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client http mountd nfs rpc-bind ssh
  ports: 2222/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
```

- The new port has been added.

-Also Restarted the ssh service as well , (systemctl restart sshd).

-Trying to connect to a remote server, with newly entered port

Task 2.2 - Configure the Apache Web Server

-Creating a directory called /companyweb.

- Coping the files from the default web server location to the /companyweb dir.

```
[root@Astra ~]#
[root@Astra ~]# mkdir /companyweb
[root@Astra ~]# ls -l /var/www/html
total 8
-rw-r--r--. 1 root root 205 Sep  9 04:21 index.html
-rw-r--r--. 1 root root  13 Sep  9 02:15 newFile.txt
[root@Astra ~]# cp /var/www/html/* /companyweb
[root@Astra ~]# ls -lZ /companyweb
total 8
-rw-r--r--. 1 root root unconfined_u:object_r:default_t:s0 205 Sep 14 00:08
-rw-r--r--. 1 root root unconfined_u:object_r:default_t:s0  13 Sep 14 00:08
[root@Astra ~]# vim /etc/httpd/
conf/           conf.d/         conf.modules.d/ logs/           modules/
[root@Astra ~]# vim /etc/httpd/conf/
httpd.conf  magic
[root@Astra ~]# vim /etc/httpd/conf/httpd.conf
```

- For the files placed inside the /companyweb to be accessible through the Apache web server , the http configuration file should be edited, and new path should be mentioned as well.
- To configure Apache to use /companyweb as the web root.

```
# symbolic links and aliases may be used to point to other locations
#
DocumentRoot "/companyweb"

#
# Relax access to content within /var/www.
#
<Directory "/var/www">
    AllowOverride None
    # Allow open access:
    Require all granted
</Directory>

# Further relax access to the default document root:
<Directory "/companyweb">
    #
    # Possible values for the Options directive are "None", "All",
    # or any combination of:
    #   Indexes Includes FollowSymLinks SymLinksifOwnerMatch ExecCGI
    #
    # Note that "MultiViews" must be named *explicitly* --- "Options
```

- /companyweb directory is specified in the configuration file.
- After editing a configuration file, it must be restarted.(systemctl restart httpd).

- We configure the SELinux config file content so that the SELinux allows Apache to access and serve files from the /companyweb directory.

```
total 4
-rw-r--r--. 1 root root unconfined_u:object_r:httpd_sys_content_t:s0 8 Sep 10 11:35 new.html
[root@Astra ~]# semanage fcontext -a -t httpd_sys_content_t "/companyweb(/.*)?"
[  674.269181] SELinux:  Converting 408 SID table entries...
[  674.281150] SELinux:  policy capability network_peer_controls=1
[  674.281319] SELinux:  policy capability open_perms=1
[  674.281447] SELinux:  policy capability extended_socket_class=1
[  674.281631] SELinux:  policy capability always_check_network=0
[  674.281773] SELinux:  policy capability cgroup_seclabel=1
[  674.281916] SELinux:  policy capability nnp_nosuid_transition=1
[  674.282039] SELinux:  policy capability genfs_seclabel_symlinks=1
[root@Astra ~]# restorecon -FR -v /companyweb
Relabeled /companyweb from unconfined_u:object_r:default_t:s0 to system_u:object_r:httpd_sys_content_t:s0
Relabeled /companyweb/index.html from unconfined_u:object_r:default_t:s0 to system_u:object_r:httpd_sys_c
Relabeled /companyweb/newFile.txt from unconfined_u:object_r:default_t:s0 to system_u:object_r:httpd_sys_
[root@Astra ~]#
```

- restorecon is used to specfiy that changes has been appiled to selinux policy.

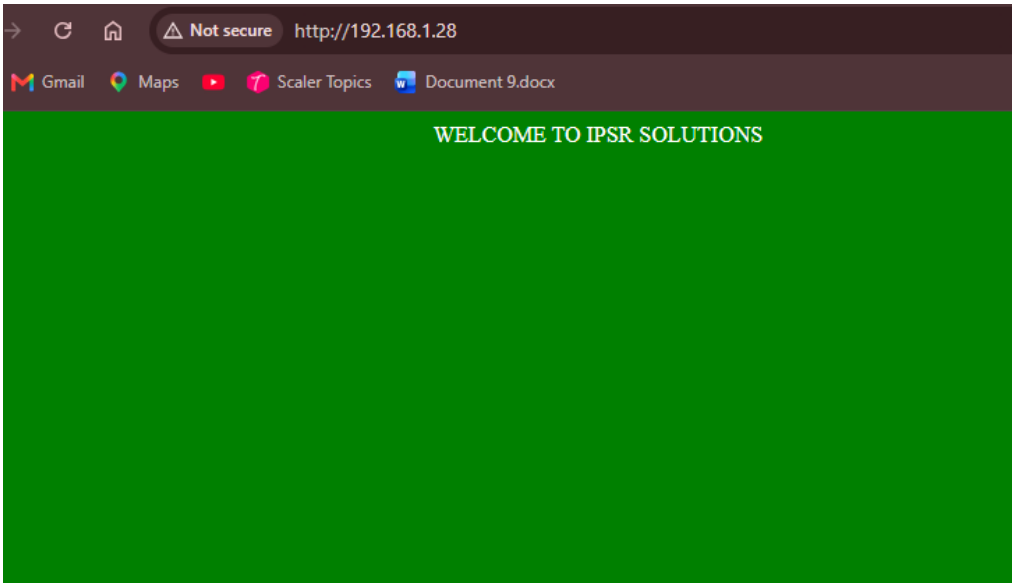
- For the Apache to be able to access and serve the content directory, check the file context. After SELinux file context configuration.

```
  rich rules:
[root@Astra ~]# ls -lZ /companyweb
total 8
-rw-r--r--. 1 root root system_u:object_r:httpd_sys_content_t:s0 205 Sep 14 00:08 index.html
-rw-r--r--. 1 root root system_u:object_r:httpd_sys_content_t:s0  13 Sep 14 00:08 newFile.txt
[root@Astra ~]#
To direct input to this VM, click inside or press Ctrl+G.
```

- It's also necessary to check if the services are fed to the firewall as well or else the http services will is blocked by the firewall. Don't forget to make it permanent.

```
[root@Astra ~]# firewall-cmd --add-service=http
success
[root@Astra ~]# firewall-cmd --list-all
public (active)
  target: default
  icmp-block-inversion: no
  interfaces: ens160
  sources:
  services: cockpit dhcpv6-client http mountd nfs rpc-bind ssh
  ports: 2222/tcp
  protocols:
  forward: yes
  masquerade: no
  forward-ports:
  source-ports:
  icmp-blocks:
  rich rules:
[root@Astra ~]# firewall-cmd --reload
success
[root@Astra ~]# firewall-cmd --add-service=http --permanent
Warning: ALREADY_ENABLED: http
success
[root@Astra ~]# firewall-cmd --list-all
```

- The configurations remains persistant after reboot. And Web page also avaliable .



Task 2.3 - Configure System Performance Profile.

- For Configuring the system use to Power Save, use tuned services. (tuned-adm profile=<profile_name>)
- To check if tuned is active or not – tuned-adm active , it will show the current profile.
- To ensure that the configuration is persistant , make sure the tuned services are enabled.

```
[root@Astra ~]# tuned-adm list
Available profiles:
- accelerator-performance     - Throughput performance based tuning with disab
- aws                         - Optimize for aws ec2 instances
- balanced                    - General non-specialized tuned profile
- balanced-battery            - Balanced profile biased towards power savings
- desktop                     - Optimize for the desktop use-case
- hpc-compute                 - Optimize for HPC compute workloads
- intel-sst                   - Configure for Intel Speed Select Base Frequenc
- latency-performance         - Optimize for deterministic performance at the
- network-latency             - Optimize for deterministic performance at the
- network-throughput          - Optimize for streaming network throughput, gen
- optimize-serial-console     - Optimize for serial console use.
- powersave                   - Optimize for low power consumption
- throughput-performance      - Broadly applicable tuning that provides excell
- virtual-guest               - Optimize for running inside a virtual guest
- virtual-host                - Optimize for running KVM guests
No current active profile.
[root@Astra ~]# tuned-adm profile powersave
[root@Astra ~]# tuned-adm active
Current active profile: powersave
[root@Astra ~]# systemctl status tuned\
>
● tuned.service - Dynamic System Tuning Daemon
     Loaded: loaded (/usr/lib/systemd/system/tuned.service; enabled; preset: e
     Active: active (running) since Sun 2026-09-13 23:43:23 EDT; 1h 38min ago
       Docs: man:tuned(8)
             man:tuned.conf(5)
             man:tuned-adm(8)
   Main PID: 821 (tuned)
      Tasks: 4 (limit: 10437)
     Memory: 17.1M (peak: 19.2M)
        CPU: 865ms
     CGroup: /system.slice/tuned.service
             └─821 /usr/bin/python3 -Es /usr/sbin/tuned -l -P
```

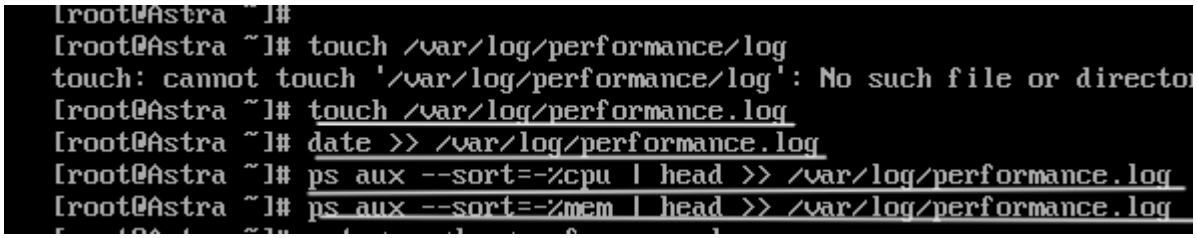
Task 2.4 -Monitor System Performance

-So the Company Wants server's CPU and Memory usage .

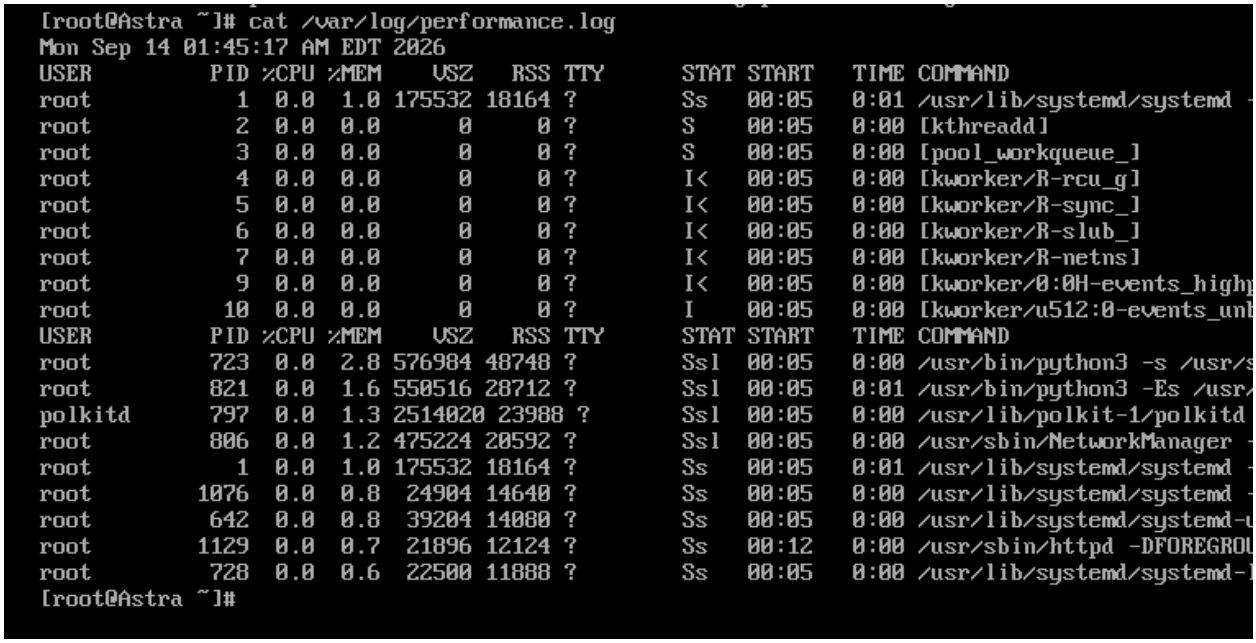
-First Create a file to store the logs about the system.[touch /var/log/performance.log]

-The ‘ps aux’ can be used to sort out the systems cpu and memory usage , so just redirect these contents to the log file.

- The Date and Time can Also be Redirected as well.



- Verify whether it’s stored in the performance.log or not.



Task 2.5 -Create a Mentoring Script

-To Create a Script file first (var/log/monitoring.sh) , then create file /var/log/system-monitor.log to save script details.

- Here is a simple script that consists of the instructions provided in the Task.

```
#!/bin/bash
echo "-----$(date '+%Y-%m-%d %H:%M:%S')-----" >> /var/log/system-monitor.log

echo "CPU USAGE:" >> /var/log/system-monitor.log
top -bn1 | grep "Cpu(s)" >>  /var/log/system-monitor.log

echo "Memory Usage: " >> /var/log/system-monitor.log
free -h >> /var/log/system-monitor.log


echo "CPU-consuming process:" >> /var/log/system-monitor.log
ps -eo pid,comm,%cpu --sort=%cpu | head -n 2 >> /var/log/system-monitor.log



echo "Memory-consuming process" >> /var/log/system-monitor.log
ps -eo pid,comm,%mem --sort=%mem | head -n 2 >> /var/log/system-monitor.log
```

-After running the script.

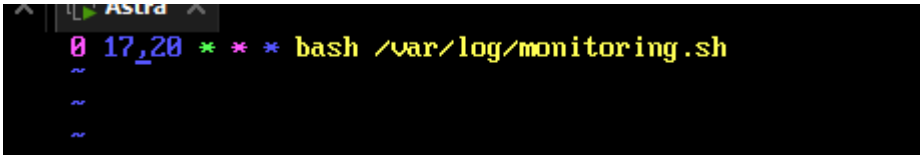
```
[root@Astra ~]# sh /var/log/monitoring.sh
[root@Astra ~]# cat /var/log/system-monitor.sh
cat: /var/log/system-monitor.sh: No such file or directory
[root@Astra ~]# cat /var/log/system-monitor.log
-----2026-09-24 12:25:33-----
CPU USAGE:
%Cpu(s):  0.0 us,  5.9 sy,  0.0 ni, 94.1 id,  0.0 wa,  0.0 hi,  0.0 si,  0.0 st
Memory Usage:
               total        used        free      shared  buff/cache   available
Mem:           1.6Gi       449Mi       1.1Gi       5.0Mi       267Mi       1.2Gi
Swap:             0B          0B          0B
CPU-consuming process:
    PID COMMAND         %CPU
      1 systemd          0.0
Memory-consuming process
    PID COMMAND         %MEM
      2 kthreadd         0.0
[root@Astra ~]# _
```

Task 2.6 - Scheduling the Monitoring Script

-We have Created a Simple Monitoring Script in the pervious task (montoring.sh).

-For this script to be executed during the mentioned period, we can schedule it with crond.

-Cron jobs are made with :crontab –e command.



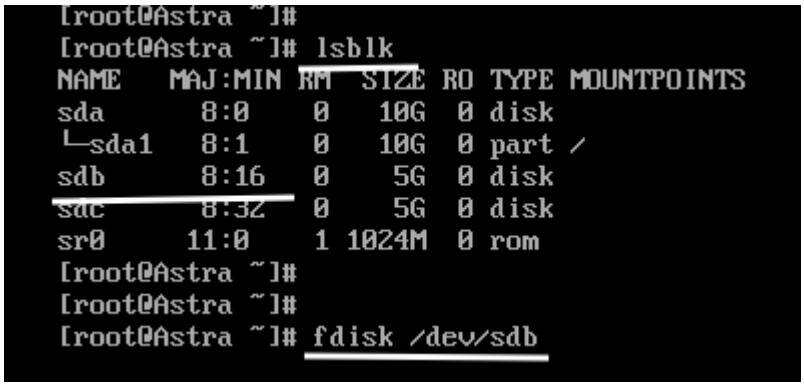
-Checking if the cronjob is persistent or not.



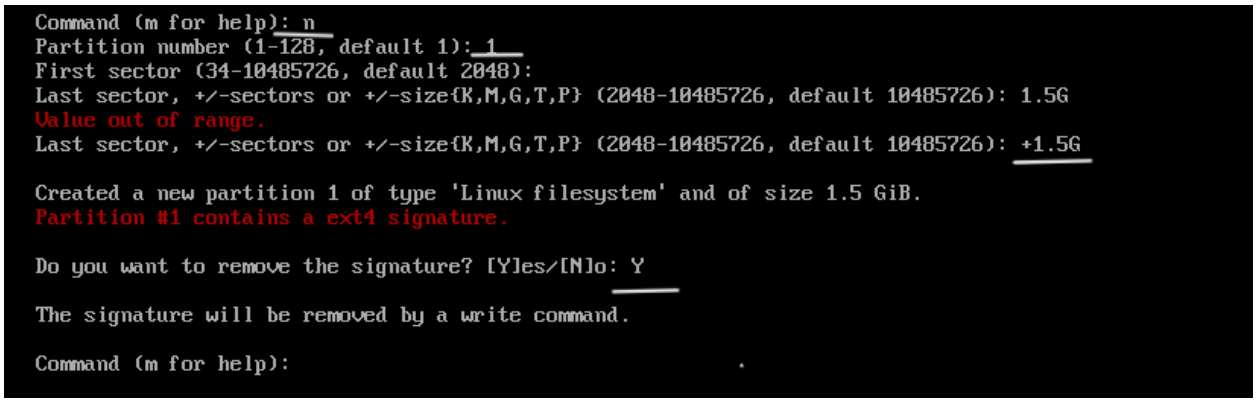
STEP 3 – STORAGE, SWAP & EXPANDABLE PARTITION

Task 3.1-Create Swap Space.

- Before Creating a 1GB swap space, its important to check whether there is space avaliable in the disk or not.



- There is 5GB free in sdb so , swap can be created there. Before creating the swap memory, a partition of that disk should be done, Here we are using MBR partitioning. -fdisk
- Inside the fdisk , specify the space needed(1.5GB in this case) and its file type.



```
Command (m for help): t
Selected partition 1
Partition type or alias (type L to list all): 19
Changed type of partition 'Linux swap' to 'Linux swap'.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
[  573.703716]  sdb: sdb1
Syncing disks.

[root@Astra ~]#
```

- After saving , it returns to the root user.
- To setup swap space use mkswap.
- The Kernal should be informed about the changes made in the disk.

```
[root@Astra ~]# partprobe
[root@Astra ~]# mkswap /dev/sdb1
Setting up swapspace version 1, size = 1.5 GiB (1610608640 bytes)
no label, UUID=df63e83c-ace3-4d74-9bba-857c068a7a88
[root@Astra ~]# partprobe
[root@Astra ~]# lsblk
```

- To Ensure that the swap space remains available after reboot, we must permanently mention it, In a configuration file “vim /etc/fstab”.

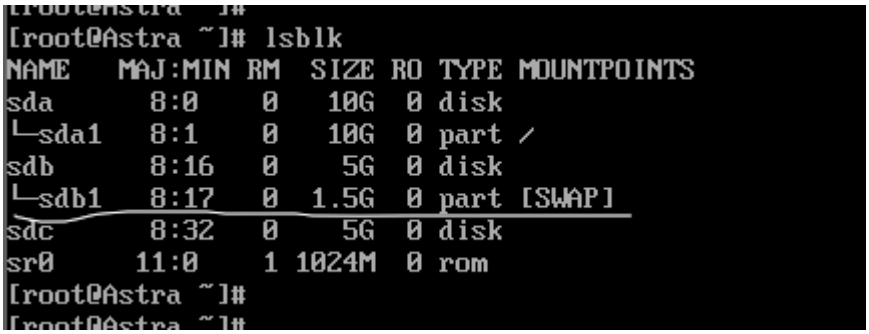
```
#
# /etc/fstab
# Created by anaconda on Sat Jan  3 14:20:00 2026
#
# Accessible filesystems, by reference, are maintai
# See man pages fstab(5), findfs(8), mount(8) and/o
#
# After editing this file, run 'systemctl daemon-re
# units generated from this file.
#
UUID=d1aa5bd2-a705-4dcf-b83a-307ce9263a6e /
/dev/sdb1 swap swap defaults 0 0
~
~
```

- For Verifying whether swap memory is formed or not ,just type lsblk in the terminal.

```
[root@Astra ~]# lsblk
NAME   MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda      8:0    0   10G  0 disk
└─sda1   8:1    0   10G  0 part /
sdb      8:16   0    5G  0 disk
└─sdb1   8:17   0  1.5G  0 part [SWAP]
sdc      8:32   0    5G  0 disk
sr0     11:0    1 1024M  0 rom
[root@Astra ~]#
```

Task 3.2 -Create an Expandable Data Storage.

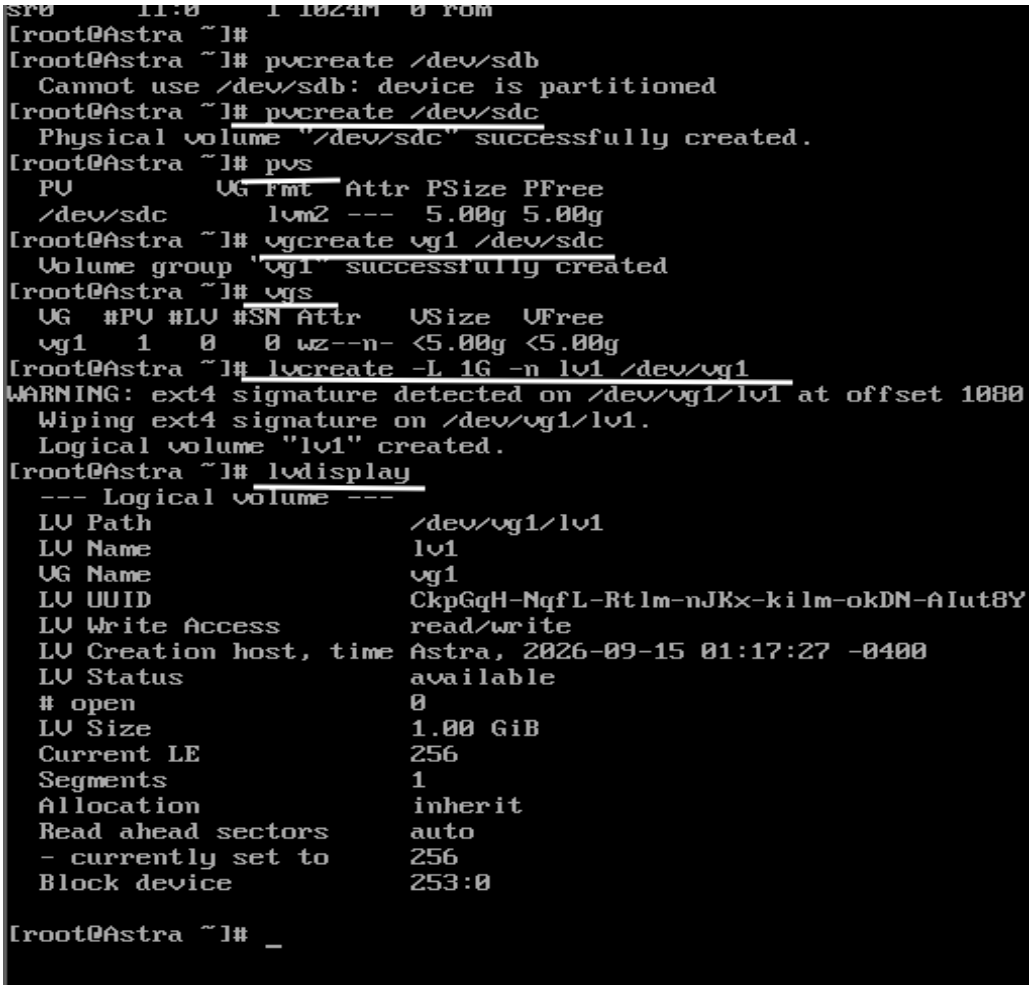
- To Create 1 GB data storage, we have to check whether the space is avalible in the disk or not.



- Here sdc , have a free space of 5GB.

- Lvm(Logical volume management) can be used to create data storage spaces.

- First create a physical volume, then a volume group on top of that creates a logical volume.



- Choosing a appropriate file system(ext4).

```
[root@Astra ~]#
[root@Astra ~]#
[root@Astra ~]# mkfs.ext4 /dev/vg1/lv1
mke2fs 1.46.5 (30-Dec-2021)
Creating filesystem with 262144 4k blocks and 65536 inodes
Filesystem UUID: 3f618d13-4bab-46e6-a8a5-cb1672168c10
Superblock backups stored on blocks:
        32768, 98304, 163840, 229376

Allocating group tables: done
Writing inode tables: done
Creating journal (8192 blocks): done
Writing superblocks and filesystem accounting information: done

[root@Astra ~]#
```

- Permanently mounting it to /mnt/company_data location, becasue the mount configuration should be persistant after reboot.

- The Configuration file vim /etc/fstab.

```
/etc/fstab
Created by anaconda on Sat Jan  3 14:20:00 2026

Accessible filesystems, by reference, are maintained
See man pages fstab(5), findfs(8), mount(8) and/or bl

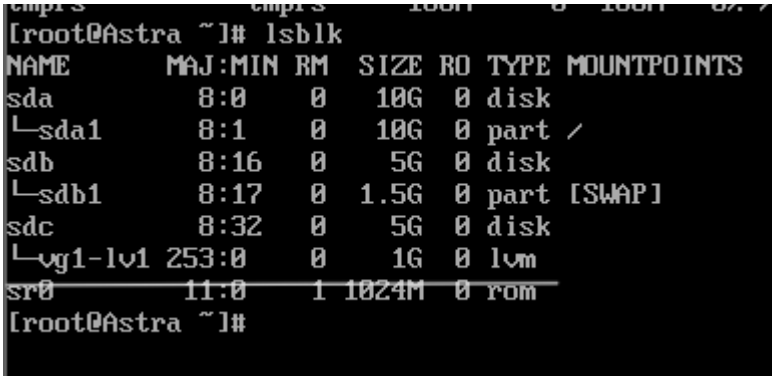
After editing this file, run 'systemctl daemon-reload
units generated from this file.

UID=d1aa5bd2-a705-4dcf-b83a-307ce9263a6e /
dev/sdb1 swap swap defaults 0 0
dev/vg1/lv1 /mnt/company_data ext4 defaults 0 0
```

- Adding Some important company data to this storage&Verifying the storage.

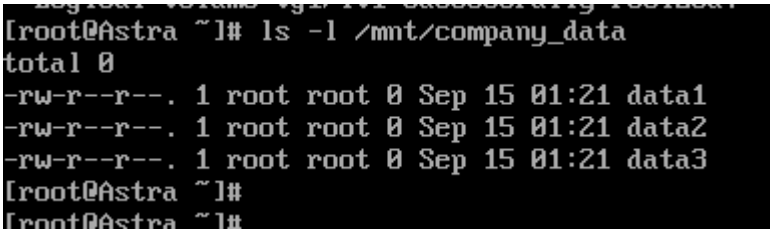
```
[root@Astra ~]# ls -l /mnt/company_data
total 0
-rw-r--r--. 1 root root 0 Sep 15 01:21 data1
-rw-r--r--. 1 root root 0 Sep 15 01:21 data2
-rw-r--r--. 1 root root 0 Sep 15 01:21 data3
[root@Astra ~]#
[root@Astra ~]#
```

- Logical volume group created sucessfully.

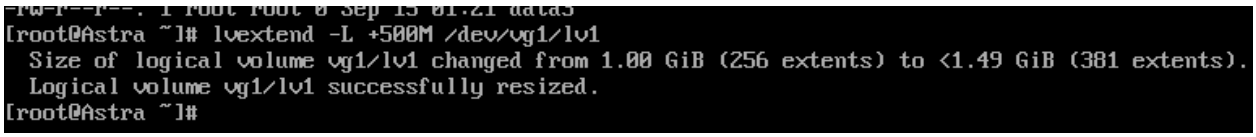


Task 3.3- Prepare the Storage for Future Expansion

- The Initial storage is set to 1GB.
- Adding sample data



- We can extend the storage using lvextend command.Suppose needs an extra 500mb storage .then,



- Resizing the file system.



STEP 4 –CREATE AN ASTRA LINUX CONTAINER IMAGE FROM THE EXISTING SYSTEM

Task 4.1 Prepare the ASTRA Linux Filesystem

-First Create a directory that will become the container root filesystem.

-Copy the filesystem into it using rsync. We don’t need things related to host kernel.

- The Astra-rootfs contain all files like system directories, Configuration Files .

```
Complete!
[root@Astra ~]# sudo rsync -aAXH --exclude=/opt/astra-rootfs --exclude
p/* --exclude=/mnt/* --exclude=/media/* --exclude=/lost+found / /opt/a
[root@Astra ~]# sudo ls -la /opt/astra-rootfs
total 40
dr-xr-xr-x. 23 root root 4096 Sep 15 10:34 .
drwxr-xr-x.  3 root root   26 Sep 18 10:39 ..
dr-xr-xr-x.  2 root root    6 Nov  2  2024 afs
drwxrwxrwt.  2 root root   25 Aug 20 00:20 Astra
lrwxrwxrwx.  1 root root    7 Nov  2  2024 bin -> usr/bin
dr-xr-xr-x.  5 root root 4096 Jan  3  2026 boot
drwxr-xr-x.  2 root root   43 Sep 14 00:24 companyweb
drwxr-xr-x.  2 root root    6 Sep 18 10:58 dev
drwxr-xr-x. 90 root root 8192 Sep 18 11:08 etc
drwxr-xr-x. 10 root root  176 Sep  7 13:00 home
drwxr-xr-x.  2 root root   91 Sep 15 11:57 Learn-bash
lrwxrwxrwx.  1 root root    7 Nov  2  2024 lib -> usr/lib
lrwxrwxrwx.  1 root root    9 Nov  2  2024 lib64 -> usr/lib64
drwxr-xr-x.  2 root root    6 Nov  2  2024 media
drwxr-xr-x.  2 root root    6 Sep 14 09:58 mnt
-rwxr-xr-x.  1 root root   50 Sep  5 00:48 myscript.sh
-rw-r--r--.  1 root root   23 Sep  9 02:14 newFile.txt
drwxr-xr-x.  2 root root    6 Sep 18 10:39 opt
dr-xr-xr-x.  2 root root    6 Sep 18 10:58 proc
dr-xr-x---.  4 root root 4096 Sep 18 10:47 root
drwxr-xr-x.  2 root root    6 Sep 18 10:58 run
lrwxrwxrwx.  1 root root    8 Nov  2  2024 sbin -> usr/sbin
-rwxrwxrwx.  1 root root  219 Aug 30 11:11 scripting
drwxr-xr-x.  3 root root   18 Sep  6 12:30 shared
drwxr-xr-x.  3 root root   22 Sep  7 13:10 srv
dr-xr-xr-x.  2 root root    6 Sep 18 10:58 sys
drwxrwxrwt.  2 root root    6 Sep 18 11:14 tmp
drwxr-xr-x. 12 root root  144 Jan  3  2026 usr
drwxr-xr-x. 19 root root 4096 Sep  8 01:01 var
drwxr-xr-x.  2 root root   22 Sep  9 12:46 website
[root@Astra ~]#
```

Task 4.2 Create the Container Image

-To Create a container first create a compressed version of the newly created directory that consists of the Linur root filesystem. (Astra-rootfs).

```
Try 'tar --help' or 'tar --usage' for more information.
[root@Astra ~]# tar --numeric-owner --xattrs --acls -czpf /tmp/astra-linux-rootfs.tar.gz -C /opt/astra-rootfs .
tar: ./var/lib/gssproxy/default.sock: socket ignored
[root@Astra ~]# ls -l /tmp/
total 1802880
-rw-r--r--. 1 root root 1846143321 Sep 18 12:25 astra-linux-rootfs.tar.gz
-rw-r--r--. 1 root root         12 Aug 29 13:02 hardlink
lrwxrwxrwx. 1 root root         19 Aug 29 13:03 softlink -> /tmp/grading/grade2
drwx------. 3 root root         17 Sep 18 10:58 systemd-private-5c50749485a3440cabcdd275ca2b43e8-chronyd.service-
drwx------. 3 root root         17 Sep 18 10:58 systemd-private-5c50749485a3440cabcdd275ca2b43e8-dbus-broker.serv
```

-Then import it to Podman . Here we are using Podman for containerization.

```
[root@Astra ~]# sudo podman import /tmp/astra-linux-rootfs.tar.gz astra-linux:latest
Getting image source signatures
Copying blob 26f3ffa9ca72 done   |
Copying config f0cf7dc571 done   |
Writing manifest to image destination
sha256:f0cf7dc571ce7331b5511887ab6a2bb2c915a5822bb737dfd6c924945a4ca4a9
[root@Astra ~]# _
```

- The compressed file is turned into container image called astra-linux:latest.

-Verify that image exists.

```
[root@Astra ~]# sudo podman images
REPOSITORY                TAG         IMAGE ID      CREATED      SIZE
localhost/astra-linux     latest      f0cf7dc571ce  2 days ago   3.09 GB
docker.io/library/httpd   2.4         17aba4293f3b  3 weeks ago  120 MB
docker.io/library/httpd   latest      17aba4293f3b  3 weeks ago  120 MB
docker.io/library/httpd   2.2         e06c3dbbfe23  8 years ago  180 MB
[root@Astra ~]# _
```

- Inspecting the images's filesystem.

```
docker.io/library/httpd  2.2         e06c3dbbfe23  8 years ago  180 MB
[root@Astra ~]# sudo podman run --rm -it astra-linux:latest /bin/bash
[ 1225.750393] bridge: filtering via arp/ip/ip6tables is no longer available by default. Update your scripts to load br_netfilter if you need this.
[ 1225.772075] podman0: port 1(veth0) entered blocking state
[ 1225.772230] podman0: port 1(veth0) entered disabled state
[ 1225.776760] veth0: entered allmulticast mode
[ 1225.777561] veth0: entered promiscuous mode
[ 1225.781008] podman0: port 1(veth0) entered blocking state
[ 1225.781140] podman0: port 1(veth0) entered forwarding state
[ 1225.843667] Warning: Deprecated Driver is detected: nft_compat will not be maintained in a future major release and may be disabled
[ 1225.844650] Warning: Deprecated Driver is detected: nft_compat_module_init will not be maintained in a future major release and may be disabled
[root@e19a2ba4a744 /]# _
```

- Now that the container is on temporarly .Verifying user and groups

```
[root@e19a2ba4a744 ~]# cat /etc/passwd | tail
kumar_intern:x:4000:4000:employee:/home/kumar_intern:/bin/bash
dbuser1:x:4001:4001::/home/dbuser1:/bin/bash
polkitd:x:996:995:User for polkitd:/:/sbin/nologin
pcp:x:994:994:Performance Co-Pilot:/var/lib/pcp:/usr/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
priya:x:4002:4002::/home/priya:/bin/bash
arjun:x:4003:4003::/home/arjun:/bin/bash
devika:x:4004:4004::/home/devika:/bin/bash
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
[root@e19a2ba4a744 ~]#
```

- Verifying installed packages

```
[root@e19a2ba4a744 ~]# ls /var/lib/rpm/
rpmdb.sqlite  rpmdb.sqlite-shm  rpmdb.sqlite-wal
[root@e19a2ba4a744 ~]#
```

Task 4.3 -Create Container Image

- Creating the container from astra-linux:latest with a name called astra-container.

```
[ 1751.911034] podman0: port 1(veth0) entered disabled state
[root@Astra ~]# sudo podman create --name astra-container astra-linux:latest /bin/bash
3bc65f6b3b15d8e8bc732e77488b4387170f6c858a0c48dcd65e4000c97ba65d
[root@Astra ~]# sudo podman ps -a
CONTAINER ID  IMAGE                              COMMAND         CREATED         STATUS                     PORTS                 NAMES
360b73ddcb66  docker.io/library/httpd:latest     /bin/bash       9 days ago      Exited (0) 9 days ago      80/tcp                myweb
15feb26b9ae9  docker.io/library/httpd:2.4        httpd-foreground  9 days ago    Exited (0) 292 years ago   0.0.0.0:8080->80/tcp  apache2.4
0c2205301387  docker.io/library/httpd:2.2        httpd-foreground  9 days ago    Exited (0) 292 years ago   0.0.0.0:8081->80/tcp  apache2.2
3bc65f6b3b15  localhost/astra-linux:latest       /bin/bash       13 seconds ago  Created                                          astra-container
[root@Astra ~]#
```

- Verifying it.

```
[root@Astra ~]# sudo podman ps
CONTAINER ID  IMAGE                           COMMAND         CREATED         STATUS         PORTS       NAMES
7286a9d9798b  localhost/astra-linux:latest    sleep infinity  34 seconds ago  Up 12 seconds              astra-container
[root@Astra ~]# _
```

-Starting the Container

```
[root@Astra ~]# sudo podman start astra-container
[ 2497.968061] podman0: port 1(veth0) entered blocking state
[ 2497.968223] podman0: port 1(veth0) entered disabled state
[ 2497.968900] veth0: entered allmulticast mode
[ 2497.969763] veth0: entered promiscuous mode
[ 2497.973718] podman0: port 1(veth0) entered blocking state
[ 2497.973874] podman0: port 1(veth0) entered forwarding state
astra-container
[root@Astra ~]# [ 2498.154580] podman0: port 1(veth0) entered disabled state
[ 2498.155623] veth0 (unregistering): left allmulticast mode
[ 2498.155823] veth0 (unregistering): left promiscuous mode
[ 2498.156050] podman0: port 1(veth0) entered disabled state
```

- Then run the container.

```
7286a9d9798b  localhost/astra-linux:latest  sleep infinity
[root@Astra ~]# sudo podman exec -it astra-container /bin/ba
[root@7286a9d9798b /]# cat /etc/os-release
NAME="Astra Linux"
VERSION="9.7 (Blue Onyx)"
ID="rocky"
ID_LIKE="rhel centos fedora"
VERSION_ID="9.7"
PLATFORM_ID="platform:el9"
PRETTY_NAME="Astra Linux 9.7 (Blue Onyx)"
ANSI_COLOR="0;32"
LOGO="fedora-logo-icon"
CPE_NAME="cpe:/o:rocky:rocky:9::baseos"
HOME_URL="https://rockylinux.org/"
VENDOR_NAME="RESF"
VENDOR_URL="https://resf.org/"
BUG_REPORT_URL="https://bugs.rockylinux.org/"
SUPPORT_END="2032-05-31"
ROCKY_SUPPORT_PRODUCT="Rocky-Linux-9"
ROCKY_SUPPORT_PRODUCT_VERSION="9.7"
REDHAT_SUPPORT_PRODUCT="Rocky Linux"
REDHAT_SUPPORT_PRODUCT_VERSION="9.7"
[root@7286a9d9798b /]# _
```

- We are inside the container, while the container is running on the same Astra Linux host.
- Verifying that the required ASTRA filesystems exit.

```
[root@7286a9d9798b /]# ls /
afs   bin   companyweb  etc   Learn-bash  lib64  mnt         newFile.txt  proc  run   scripting  srv  tmp  var
Astra  boot  dev        home  lib         media  myscript.sh  opt          root  sbin  shared     sys  usr  website
[root@7286a9d9798b /]# _
```

-Verifying Groups.

```
[root@7286a9d9798b /]# cat /etc/group | tail -n -5
priya:x:4002:
arjun:x:4003:
devika:x:4004:
devteam:x:50001:priya,arjun,devika
apache:x:48:
```

Task 4.4 - Configure SSH Inside the Container

- In Task 4.3, we created a container(astra-container), that container has no port number.

- Stop and remove the old container

```
[root@Astra ~]# sudo podman stop astra-container
WARN[0010] StopSignal SIGTERM failed to stop container astra-container in 10 seconds, resorting to SIGKILL
[ 4454.861661] podman0: port 1(veth0) entered disabled state
[ 4454.862701] veth0 (unregistering): left allmulticast mode
[ 4454.862849] veth0 (unregistering): left promiscuous mode
[ 4454.863002] podman0: port 1(veth0) entered disabled state
astra-container
[root@Astra ~]# sudo podman rm astra-container
astra-container
[root@Astra ~]# sudo podman ps -a
CONTAINER ID  IMAGE                              COMMAND          CREATED     STATUS                   PORTS                   NAMES
360b73ddcb66  docker.io/library/httpd:latest     /bin/bash        9 days ago  Exited (0) 9 days ago    80/tcp                  myweb
15feb26b9ae9  docker.io/library/httpd:2.4        httpd-foreground  9 days ago  Exited (0) 292 years ago  0.0.0.0:8080->80/tcp  apache2.4
0c2205301387  docker.io/library/httpd:2.2        httpd-foreground  9 days ago  Exited (0) 292 years ago  0.0.0.0:8081->80/tcp  apache2.2
[root@Astra ~]# _
```

  - Create the container with SSH port mapping and Start it.

```
[root@Astra ~]#
[root@Astra ~]# sudo podman create --name astra-container -p 2222:22  astra-linux:latest sleep infinity
d85769ada542d5ce7d9fe78dbe21359210421aacb0cfb7161337f3cdc9fe5a24
[root@Astra ~]# sudo podman start astra-container
[ 5172.036965] podman0: port 1(veth0) entered blocking state
[ 5172.037407] podman0: port 1(veth0) entered disabled state
[ 5172.041055] veth0: entered allmulticast mode
[ 5172.042167] veth0: entered promiscuous mode
[ 5172.046408] podman0: port 1(veth0) entered blocking state
[ 5172.046746] podman0: port 1(veth0) entered forwarding state
astra-container
[root@Astra ~]# sudo podman ps
CONTAINER ID  IMAGE                          COMMAND         CREATED        STATUS        PORTS                 N
d85769ada542  localhost/astra-linux:latest  sleep infinity  2 minutes ago  Up 2 minutes  0.0.0.0:2222->22/tcp  a
[root@Astra ~]# _
```

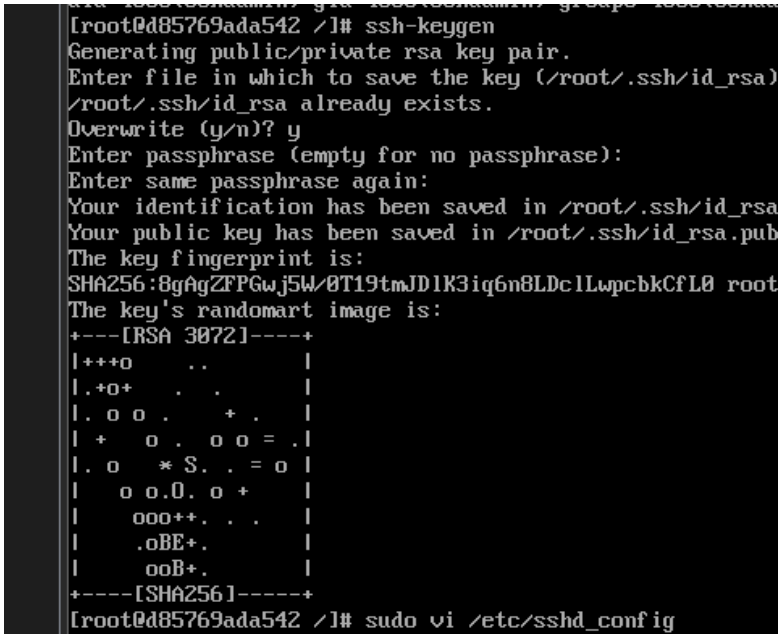
- Once Inside the container , check whether the SSH packages are installed or not.

```
root@d85769ada542 /]#
root@d85769ada542 /]# rpm -q openssh-clients
penssh-clients-8.7p1-46.el9.rocky.0.1.x86_64
root@d85769ada542 /]# rpm -q openssh-server
penssh-server-8.7p1-46.el9.rocky.0.1.x86_64
root@d85769ada542 /]#
```

  - Creating an SSH user.

```
[root@d85769ada542 /]# useradd sshadmin
[root@d85769ada542 /]# passwd sshadmin
Changing password for user sshadmin.
New password:
BAD PASSWORD: The password is shorter than 8 characters
Retype new password:
passwd: all authentication tokens updated successfully.
[root@d85769ada542 /]# id sshadmin
uid=4005(sshadmin) gid=4005(sshadmin) groups=4005(sshadmin)
[root@d85769ada542 /]# _
```

    - Configuring SSH , First creating a SSH host key, we are using key based authentication method inside the container.

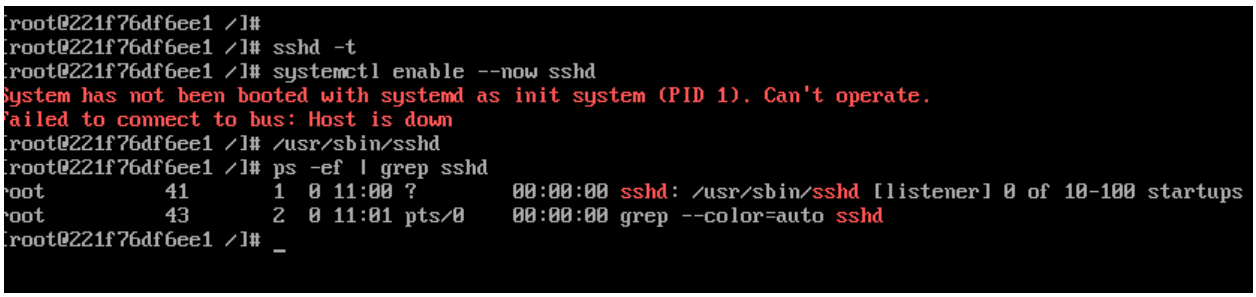


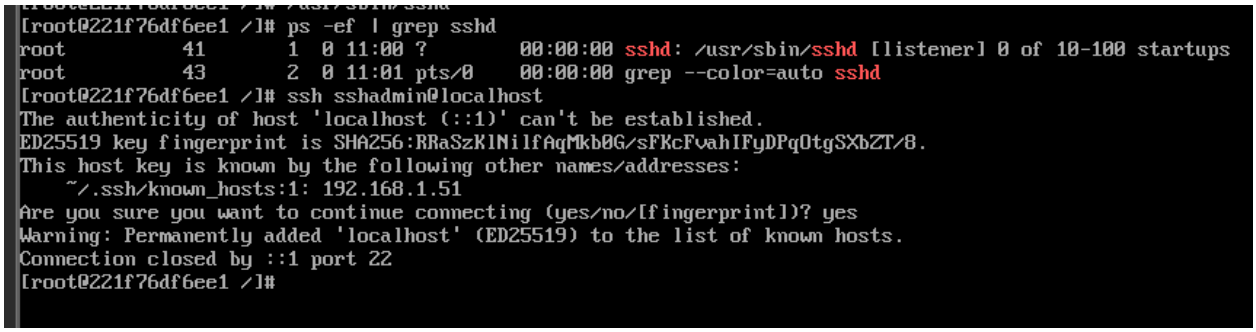
- Editing the configuration file /etc/ssh/sshd_config and Ensuring these lines are set Port 22, PermitRootLogin ,PasswordAuthentication yes .

- Enabling the ssh (This is done from another container, the steps followed were the same.)

- Start SSH inside the container(usr/sbin/ssh).

- Verifying that SSH services are available and working correctly from inside the container.





- It Accepted the host key then its showing the connection is closed , so we are not being able to connect .

- Now lets configure this and find what was causing the problem.
- To figure out where the problem is Kill all sshd forcefully.Then start the sshd , redirect the output to a log file to inspect what went wrong.



- The Log file is Showing “linux_audit_write_entry failed: Operation not permitted”.

- So After a little research, what this error means is that sshd tries to write a login audit record to the Linux kernel's audit subsystem via a netlink socket. This requires the CAP_AUDIT_WRITE capability, which Podman does **not** grant to containers by default . So we to add the AUDIT_WRITE capability to the container.



- After the adding the audit file ,run once again.There is still some issue’s we still cannot establish connection.

```
91daf98f69605009bed0272a2794d1afb84fca6a753c76cfb8359bfac394e84
[root@Astra ~]# podman exec -it astra-container /bin/bash
[root@91daf98f6960 /]#
[root@91daf98f6960 /]# usr/sbin/sshd
sshd re-exec requires execution with an absolute path
[root@91daf98f6960 /]# usr/sbin/sshd
sshd re-exec requires execution with an absolute path
[root@91daf98f6960 /]# cd
[root@91daf98f6960 ~]# usr/sbin/sshd
bash: usr/sbin/sshd: No such file or directory
[root@91daf98f6960 ~]# /usr/sbin/sshd
[root@91daf98f6960 ~]# ssh sshadmin@localhost
sshadmin@localhost's password:
Connection to localhost closed.
[root@91daf98f6960 ~]#
To direct input to this VM, click inside or press Ctrl+G.
```

- So once again kill all sshd process and start it again ,then redirect the output to log file.

```
[root@91daf98f6960 ~]# cat /tmp/sshd_debug.log
Server listening on 0.0.0.0 port 22.
Server listening on :: port 22.
Accepted password for sshadmin from ::1 port 60028 ssh2
PAM: pam_open_session(): Cannot make/remove an entry for the specified session
syslogin_perform_logout: logout() returned an error
Received disconnect from ::1 port 60028:11: disconnected by user
Disconnected from user sshadmin ::1 port 60028
[root@91daf98f6960 ~]# sshd -T | grep -i userpam
[root@91daf98f6960 ~]# cat /etc/ssh/sshd_config | grep -i include
#  /etc/ssh/sshd_config.d/  which will be automatically included below
Include /etc/ssh/sshd_config.d/*.conf
[root@91daf98f6960 ~]# ls -la /etc/ssh/sshd_config.d/
total 8
drwx------. 2 root root  59 Jan  3  2026 .
drwxr-xr-x. 1 root root  25 Sep 24 02:20 ..
-rw-r--r--. 1 root root 141 Jan  3  2026 01-permitrootlogin.conf
-rw-------. 1 root root 719 Nov  3  2025 50-redhat.conf
[root@91daf98f6960 ~]# grep -i usepam /etc/ssh/sshd_config.d/*.conf
/etc/ssh/sshd_config.d/50-redhat.conf:UsePAM yes
[root@91daf98f6960 ~]# vi /etc/ssh/sshd_config.d/50-redhat.conf
```

- Here PAM fails to create the login session, so SSH disconnects immediately after successful authentication.

- To solve this we can set usePAM to no inside sshd configuration file.50-redhat.conf file was overriding my usePAM no change in the sshd confihuration file.

- After setting usePAM to no , restarted sshd.

```
GSSAPIAuthentication yes
GSSAPICleanupCredentials no

UsePAM no_

X11Forwarding yes
```

- Finally ssh connection established, verified with whoami .

```
"/etc/ssh/sshd_config.d/50-redhat.conf" 22L, 718B written
[root@91daf98f6960 ~]# sshd -T | grep -i usepam
WARNING: 'UsePAM no' is not supported in RHEL and may cause several problems.
usepam no
[root@91daf98f6960 ~]# pkill -9 sshd
[root@91daf98f6960 ~]# /usr/sbin/sshd -D -e > /tmp/sshd_debug.log 2>&1 &
[1] 134
[root@91daf98f6960 ~]# ssh sshadmin@localhost
sshadmin@localhost's password:
Last login: Thu Sep 24 02:21:02 2026 from ::1
[sshadmin@91daf98f6960 ~]$ whomai
-bash: whomai: command not found
[sshadmin@91daf98f6960 ~]$ whoami
sshadmin
[sshadmin@91daf98f6960 ~]$
```

Task 4.5-Configure Port Mapping

- Save current work as an image.

```
[root@Astra ~]# sudo podman commit astra-container astra-linux:ssh
Getting image source signatures
Copying blob f277a4a1e1a2 skipped: already exists
Copying blob bbb0ff7f1ad1 skipped: already exists
Copying blob 4041bc1bf0e6 done   |
Copying config 75c26c3ac4 done   |
Writing manifest to image destination
75c26c3ac4d5e2fbedf87e7de41e2afac27a9be4fd07189d35dd698db92e1dbc
[root@Astra ~]# sudo podman images
REPOSITORY                  TAG         IMAGE ID      CREATED             SIZE
localhost/astra-linux       ssh         75c26c3ac4d5  17 seconds ago      3.09 GB
localhost/astra-cont-image  latest      40d582103ff3  About an hour ago   3.09 GB
<none>                      <none>      81edc832ff72  2 days ago          3.09 GB
<none>                      <none>      dea88da27011  2 days ago          3.09 GB
localhost/astra-linux       latest      f0cf7dc571ce  5 days ago          3.09 GB
docker.io/library/httpd     2.4         17aba4293f3b  4 weeks ago         120 MB
docker.io/library/httpd     latest      17aba4293f3b  4 weeks ago         120 MB
docker.io/library/httpd     2.2         e06c3dbbfe23  8 years ago         180 MB
```

- Then, Remove the old container.

```
[root@Astra ~]# sudo podman stop astra-container
WARN[0010] StopSignal SIGTERM failed to stop container astra-con
[10312.549477] podman0: port 1(veth0) entered disabled state
[10312.554010] veth0 (unregistering): left allmulticast mode
[10312.554279] veth0 (unregistering): left promiscuous mode
[10312.554580] podman0: port 1(veth0) entered disabled state
astra-container
[root@Astra ~]# sudo podman rm astra-container
astra-container
[root@Astra ~]# _
```

- Run a new container with the correct port mapping 4444 to 22

```
[root@Astra ~]# sudo podman run -dit --name astra-container -p 4444:22 --cap-add=AUDIT_WRITE astra-linux:ssh tail -f /dev/null
[10548.822114] podman0: port 1(veth0) entered blocking state
[10548.822362] podman0: port 1(veth0) entered disabled state
[10548.823212] veth0: entered allmulticast mode
[10548.824040] veth0: entered promiscuous mode
[10548.828096] podman0: port 1(veth0) entered blocking state
[10548.828327] podman0: port 1(veth0) entered forwarding state
a933e27b78a813ad0dff34dbcc5bca251bf755ecbe3b8540f4081998d6989721
[root@Astra ~]#
```

- Start SSH inside the container.

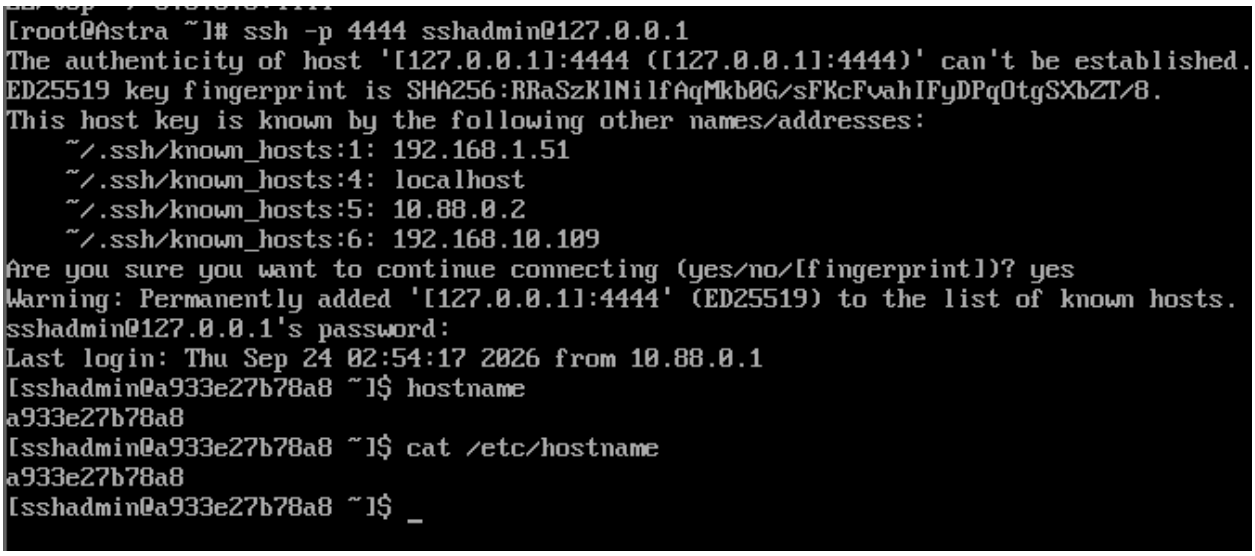
```
[root@Astra ~]# podman exec -it astra-container bash
[root@a933e27b78a8 /]# cd
[root@a933e27b78a8 ~]# /usr/sbin/sshd
WARNING: 'UsePAM no' is not supported in RHEL and may cause several problems.
[root@a933e27b78a8 ~]# exit
exit
```

- Verify that the port is active

```
[root@Astra ~]# podman port astra-container
22/tcp -> 0.0.0.0:4444
[root@Astra ~]# _
```

Task 4.6 - Access the ASTRA Container Through SSH

-Connect from the host using port 4444.



-Verify the connection that reaches the container, not the host.

```
a933e27b78a8
[sshadmin@a933e27b78a8 ~]$ cat /etc/os-release
NAME="Astra Linux"
VERSION="9.7 (Blue Onyx)"
ID="rocky"
ID_LIKE="rhel centos fedora"
VERSION_ID="9.7"
PLATFORM_ID="platform:el9"
PRETTY_NAME="Astra Linux 9.7 (Blue Onyx)"
ANSI_COLOR="0;32"
LOGO="fedora-logo-icon"
CPE_NAME="cpe:/o:rocky:rocky:9::baseos"
HOME_URL="https://rockylinux.org/"
VENDOR_NAME="RESF"
VENDOR_URL="https://resf.org/"
BUG_REPORT_URL="https://bugs.rockylinux.org/"
SUPPORT_END="2032-05-31"
ROCKY_SUPPORT_PRODUCT="Rocky-Linux-9"
ROCKY_SUPPORT_PRODUCT_VERSION="9.7"
REDHAT_SUPPORT_PRODUCT="Rocky Linux"
REDHAT_SUPPORT_PRODUCT_VERSION="9.7"
[sshadmin@a933e27b78a8 ~]$ _
```

- Verifying required Astra Linux files/directories exist.(ls -la).

- verifying required packages/config. (yum list --installed).

```
rsyslog.x86_64                                  8.2506.0-2.el9
rsyslog-logrotate.x86_64                        8.2506.0-2.el9
samba-client-libs.x86_64                        4.22.4-6.el9
samba-common.noarch                             4.22.4-6.el9
samba-common-libs.x86_64                        4.22.4-6.el9
sed.x86_64                                      4.8-9.el9
selinux-policy.noarch                           38.1.75-2.el9_8
selinux-policy-targeted.noarch                  38.1.75-2.el9_8
setup.noarch                                    2.13.7-10.el9
sg3_utils.x86_64                                1.47-10.el9
sg3_utils-libs.x86_64                           1.47-10.el9
shadow-utils.x86_64                             2:4.9-15.el9
shadow-utils-subid.x86_64                       2:4.9-16.el9
skopeo.x86_64                                   2:1.22.2-7.el9_8
slang.x86_64                                    2.3.2-11.el9
slirp4netns.x86_64                              1.3.3-1.el9
sqlite-libs.x86_64                              3.34.1-9.el9_7
sssd-client.x86_64                              2.9.8-4.el9_8.1
sssd-common.x86_64                              2.9.8-4.el9_8.1
sssd-kcm.x86_64                                 2.9.8-4.el9_8.1
sssd-nfs-idmap.x86_64                           2.9.8-4.el9_8.1
sudo.x86_64                                     1.9.5p2-13.el9
systemd.x86_64                                  252-55.el9_7.2.rocky
systemd-libs.x86_64                             252-55.el9_7.2.rocky
systemd-pam.x86_64                              252-55.el9_7.2.rocky
systemd-rpm-macros.noarch                       252-55.el9_7.2.rocky
systemd-udev.x86_64                             252-55.el9_7.2.rocky
tar.x86_64                                      2:1.34-11.el9
teamd.x86_64                                    1.31-16.el9_1
toolbox.x86_64                                  0.3-2.el9_8.rocky.0
```

Task 4.7 Verify the ASTRA Linux Environment

-Operating system information.

```
a933e27b78a8
[sshadmin@a933e27b78a8 ~]$ cat /etc/os-release
NAME="Astra Linux"
VERSION="9.7 (Blue Onyx)"
ID="rocky"
ID_LIKE="rhel centos fedora"
VERSION_ID="9.7"
PLATFORM_ID="platform:el9"
PRETTY_NAME="Astra Linux 9.7 (Blue Onyx)"
ANSI_COLOR="0;32"
LOGO="fedora-logo-icon"
CPE_NAME="cpe:/o:rocky:rocky:9::baseos"
HOME_URL="https://rockylinux.org/"
VENDOR_NAME="RESF"
VENDOR_URL="https://resf.org/"
BUG_REPORT_URL="https://bugs.rockylinux.org/"
SUPPORT_END="2032-05-31"
ROCKY_SUPPORT_PRODUCT="Rocky-Linux-9"
ROCKY_SUPPORT_PRODUCT_VERSION="9.7"
REDHAT_SUPPORT_PRODUCT="Rocky Linux"
REDHAT_SUPPORT_PRODUCT_VERSION="9.7"
[sshadmin@a933e27b78a8 ~]$ _
```

- Hostname/container identity.

```
sshadmin@127.0.0.1's password:
Last login: Thu Sep 24 02:54:17 2026 from 10.88.0.1
[sshadmin@a933e27b78a8 ~]$ hostname
a933e27b78a8
[sshadmin@a933e27b78a8 ~]$ cat /etc/hostname
a933e27b78a8
[sshadmin@a933e27b78a8 ~]$ _
```

- verifying required packages/config. (yum list --installed)

```
rsyslog.x86_64                                  8.2506.0-2.el9
rsyslog-logrotate.x86_64                        8.2506.0-2.el9
samba-client-libs.x86_64                        4.22.4-6.el9
samba-common.noarch                             4.22.4-6.el9
samba-common-libs.x86_64                        4.22.4-6.el9
sed.x86_64                                      4.8-9.el9
selinux-policy.noarch                           38.1.75-2.el9_8
selinux-policy-targeted.noarch                  38.1.75-2.el9_8
setup.noarch                                    2.13.7-10.el9
sg3_utils.x86_64                                1.47-10.el9
sg3_utils-libs.x86_64                           1.47-10.el9
shadow-utils.x86_64                             2:4.9-15.el9
shadow-utils-subid.x86_64                       2:4.9-16.el9
skopeo.x86_64                                   2:1.22.2-7.el9_8
slang.x86_64                                    2.3.2-11.el9
slirp4netns.x86_64                              1.3.3-1.el9
sqlite-libs.x86_64                              3.34.1-9.el9_7
sssd-client.x86_64                              2.9.8-4.el9_8.1
sssd-common.x86_64                              2.9.8-4.el9_8.1
sssd-kcm.x86_64                                 2.9.8-4.el9_8.1
sssd-nfs-idmap.x86_64                           2.9.8-4.el9_8.1
sudo.x86_64                                     1.9.5p2-13.el9
systemd.x86_64                                  252-55.el9_7.2.rocky
systemd-libs.x86_64                             252-55.el9_7.2.rocky
systemd-pam.x86_64                              252-55.el9_7.2.rocky
systemd-rpm-macros.noarch                       252-55.el9_7.2.rocky
systemd-udev.x86_64                             252-55.el9_7.2.rocky
tar.x86_64                                      2:1.34-11.el9
teamd.x86_64                                    1.31-16.el9_1
toolbox.x86_64                                  0.3-2.el9_8.rocky.0
```

- Required Users and Groups

```
[sshadmin@a933e27b78a8 ~]$ cat /etc/group | tail
polkitd:x:995:
pcp:x:994:
rpc:x:32:
rpcuser:x:29:
priya:x:4002:
arjun:x:4003:
devika:x:4004:
devteam:x:50001:priya,arjun,devika
apache:x:48:
sshadmin:x:4005:
[sshadmin@a933e27b78a8 ~]$ cat /etc/passwd | tail
dbuser1:x:4001:4001::/home/dbuser1:/bin/bash
polkitd:x:996:995:User for polkitd:/:/sbin/nologin
pcp:x:994:994:Performance Co-Pilot:/var/lib/pcp:/usr/sbin/nologin
rpc:x:32:32:Rpcbind Daemon:/var/lib/rpcbind:/sbin/nologin
rpcuser:x:29:29:RPC Service User:/var/lib/nfs:/sbin/nologin
priya:x:4002:4002::/home/priya:/bin/bash
arjun:x:4003:4003::/home/arjun:/bin/bash
devika:x:4004:4004::/home/devika:/bin/bash
apache:x:48:48:Apache:/usr/share/httpd:/sbin/nologin
sshadmin:x:4005:4005::/home/sshadmin:/bin/bash
[sshadmin@a933e27b78a8 ~]$ _
```

- Important Configuration Files

```
[sshadmin@a933e27b78a8 ~]$ ls -l /etc/ssh/sshd_config
-rw-------. 1 root root 3651 Sep 24 02:20 /etc/ssh/sshd_config
[sshadmin@a933e27b78a8 ~]$ ls -l /etc/ssh/sshd_config.d
ls: cannot open directory '/etc/ssh/sshd_config.d': Permission denied
[sshadmin@a933e27b78a8 ~]$ ls -l /etc/passwd /etc/shadow
-rw-r--r--. 1 root root 1555 Sep 22 10:57 /etc/passwd
----------. 1 root root 1387 Sep 22 10:57 /etc/shadow
[sshadmin@a933e27b78a8 ~]$ _
```

- Custom Directories

```
[sshadmin@a933e27b78a8 ~]$ ls -la /home/sshadmin
total 16
drwx------. 1 sshadmin sshadmin  53 Sep 24 03:17 .
drwxr-xr-x. 1 root     root      22 Sep 22 10:57 ..
-rw-------. 1 sshadmin sshadmin 223 Sep 24 03:08 .bash_history
-rw-r--r--. 1 sshadmin sshadmin  18 Apr 30  2024 .bash_logout
-rw-r--r--. 1 sshadmin sshadmin 141 Apr 30  2024 .bash_profile
-rw-r--r--. 1 sshadmin sshadmin 492 Apr 30  2024 .bashrc
drwxr-xr-x. 2 sshadmin sshadmin   6 Sep 24 03:17 data1
drwxr-xr-x. 2 sshadmin sshadmin   6 Sep 24 03:17 data2
[sshadmin@a933e27b78a8 ~]$
```

- **Application Files**

```
[sshadmin@a933e27b78a8 ~]$ which sshd
/usr/sbin/sshd
[sshadmin@a933e27b78a8 ~]$ ls -l /usr/sbin/sshd
-rwxr-xr-x. 1 root root 968784 Nov  3  2025 /usr/sbin/sshd
[sshadmin@a933e27b78a8 ~]$ _
```

-Required Enironment settings.



THANK YOU :)